Target's Pregnancy Prediction

Predictive Analytics & Privacy Ethics

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Algorithm Basics and BW

Target's Pregnancy Prediction

Predictive Analytics & Privacy Ethics

When retail analytics crosses the privacy line

Learning Objectives

1. How predictive analytics uses purchase patterns to infer personal information
2. $ Business value of behavioral targeting and customer lifetime value
3. Privacy implications of predicting life events without consent
4. The "creep factor" and how to manage prediction transparency
5. GDPR/CCPA compliance considerations for predictive models

Key Question: Where's the line between personalization and invasion of privacy?

Background: Retail Analytics in the 2000s

Traditional marketing:

- Mass mailings
- Newspaper inserts
- TV commercials
- Low response rates (1-2%)
- $ Expensive waste
- No personalization

The opportunity:

- Transaction data available
- Loyalty cards track purchases
- Can we predict behavior?

Background: Why Target Pregnant Women?

Why target pregnant women?

- New parents = high-value customers
- Spend 60% more
- Shopping habits change
- Brand switching opportunity
- Early acquisition = lifetime loyalty

The challenge:

- Women don't announce pregnancy
- Baby registries only in 2nd trimester
- Need to identify EARLY

Target's Data Infrastructure

Data collection:

- Guest ID system
  - Tracks every purchase
  - Links credit card, email, phone
- Point-of-sale transactions
- Web browsing behavior
- Email engagement
- Baby registry data
- Third-party data purchases

What they knew:

- Demographics (age, location)
- Purchase history
- Shopping frequency
- Spending patterns
- Seasonal trends
- Store visit patterns

Key insight: Every transaction reveals something about life circumstances

The Pregnancy Prediction Project

Project leader:

- Andrew Pole, Senior Statistician
- Goal: Predict pregnancy
- Timeline: Early 2000s
- Using machine learning

The hypothesis:

- Pregnant women buy specific products
- Purchase patterns change by trimester
- Combinations more predictive than individual items
- Can identify pregnancy early (1st trimester)

The Pregnancy Prediction Project

Training data source:

- Baby registry customers
  - Known pregnant
  - Know due dates
  - Can look backward
- Thousands of examples
- High-quality labels

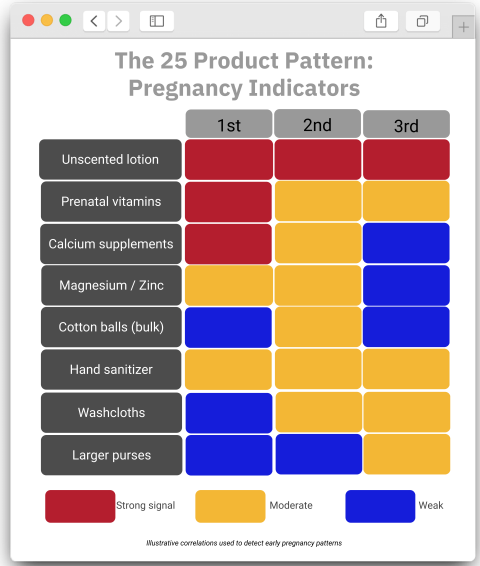
The approach:

- Supervised machine learning
- Identify product patterns
- Test on new customers
- Refine model iteratively

The 25 Product Pattern

Key products identified:

- Unscented lotion
  - Scent sensitivity in pregnancy
- Supplements
  - Calcium, magnesium, zinc
  - Prenatal vitamins
- Cotton balls (large quantities)
- Hand sanitizer
- Washcloths
- Larger purses



Predictive Model Development - Parameterization

Predictive Score for pregnancy from model:

$$
\begin{aligned}
z = \alpha & \\
& + \beta_1 \text{ Unscented Lotion (Yes} = 1 \text{ / No} = 0) \\
& + \beta_2 \text{ Prenatal Vitamins (Yes} = 1 \text{ / No} = 0) \\
& + \beta_3 \text{ Calcium Supplements (Yes} = 1 \text{ / No} = 0) \\
& + \epsilon
\end{aligned}
$$

$$
\textit{Prob}(\text{Pregnancy}) = \frac{1}{1 + e^{-z}}
$$

Beyond Individual Products: Pattern Recognition (Why combinations matter)

Single products:

- ✖ Low predictive power
- ? Many false positives
- Example: Lotion purchase
  - Could be: pregnant, dry skin, gift
  - Not enough information

Product combinations:

- ✔ High predictive power
- Example: Unscented lotion + calcium + zinc + cotton balls
- Probability: 87% pregnant

Predictive Model Development

Predictive Score for pregnancy from model based on combo:

$$
\begin{aligned}
z = \alpha & \\
& + \beta_1 \text{ Unscented Lotion (Yes} = 1 \text{ / No} = 0) \\
& + \beta_2 \text{ Prenatal Vitamins (Yes} = 1 \text{ / No} = 0) \\
& + \beta_3 \text{ Calcium Supplements (Yes} = 1 \text{ / No} = 0) \\
& + \beta_4 \text{ Unscented Lotion \& Prenatal Vitamins (Yes} = 1 \text{ / No} = 0) \\
& + \beta_5 \text{ Prenatal Vitamins \& Calcium Supplements (Yes} = 1 \text{ / No} = 0) \\
& + \beta_6 \text{ Calcium Supplements \& Unscented Lotion (Yes} = 1 \text{ / No} = 0) \\
& + \beta_7 \text{ All 3 items (Yes} = 1 \text{ / No} = 0) \\
& + \epsilon
\end{aligned}
$$

Predictive Model Development

Predictive Score for pregnancy from model based on combo and timing:

$$
\begin{aligned}
z = \alpha &+ \beta_1 \text{ Unscented Lotion (Yes} = 1 \text{ / No} = 0) \\
&+ \beta_2 \text{ Prenatal Vitamins (Yes} = 1 \text{ / No} = 0) \\
&+ \beta_3 \text{ Calcium Supplements (Yes} = 1 \text{ / No} = 0) \\
&+ \beta_4 \text{ Unscented Lotion \& Prenatal Vitamins (Yes} = 1 \text{ / No} = 0) \\
&+ \beta_5 \text{ Prenatal Vitamins \& Calcium Supplements (Yes} = 1 \text{ / No} = 0) \\
&+ \beta_6 \text{ Calcium Supplements \& Unscented Lotion (Yes} = 1 \text{ / No} = 0) \\
&+ \beta_7 \text{ All 3 items (Yes} = 1 \text{ / No} = 0) \\
&+ \beta_8 \text{ Purchase Sequence dummy (Yes} = 1 \text{ / No} = 0) \\
&+ \epsilon
\end{aligned}
$$

$$
\textit{Prob}(\text{Pregnancy}) = \frac{1}{1 + e^{-z}}
$$

Beyond Individual Products: Pattern Recognition (Why combinations matter)

Timing patterns:

- Purchase sequence matters
- Frequency changes
- Week 1-12: Supplements
- Week 13-26: Maternity clothes
- Week 27-40: Baby items

Prediction accuracy:

- Can predict due date within small window
- Can identify trimester
- 25 products analyzed together
- Sophisticated scoring algorithm

Key insight: Context + patterns + timing = powerful predictions

Business Value: How the revenue model works: $600M Annually

Customer lifetime value:

- Average customer: $50/month
- New parent: $80/month (60% increase)
- Duration: 18+ years
- Lifetime value:
  - $80 × 12 × 18 = $17,280
- ↑ vs regular customer: $10,800
- $ Difference: $6,480 per parent

The $600M calculation:

- 4M births/year in US
- Target captures 2.5% = 100K
- Early targeting = 30% higher retention
- 30K additional customers × $6,480
- → = $194M in lifetime value
- + Plus immediate pregnancy spending
- $ Total: $600M+ annually

Key driver: Acquiring customers BEFORE competitors do

Marketing Efficiency Gains

Traditional marketing:

- Mass mailing to everyone
- $1 per mailer
- 1-2% response rate
- Cost per acquisition:
  - $1 / 0.015 = $67
- 98% waste

Targeted marketing:

- Mail only to high-probability pregnant women
- 25% response rate
- $1 / 0.25 = $4 cost per acquisition

The Famous Story: Target Knew First (The incident that made headlines)

What happened:

- Target mails baby coupons
- To teenage girl's home
- Father sees coupons, complains
  "She's still in high school!"
- Store manager apologizes
- Few weeks later...
- Father apologizes to Target
  - Daughter was pregnant
  - She hadn't told him
- **Target knew before he did**

Target's Response: The Camouflage Strategy

The problem:

- ⚠ Customers felt "spied on"
- 😠 Backlash hurting brand
- ❓ But predictions were valuable
- ⚖ How to use data without creeping people out?

The solution:

- 🕶 Camouflage baby products
- 🛒 Mix with random items
- Example mailer:
  - Diapers + wine glasses
  - Baby lotion + lawn mower
  - Crib sheets + power tools

Target's Response: The Camouflage Strategy

The psychology:

- Looks like general catalog
- Pregnant women notice baby items
- Others ignore them
- Reduces "creep factor"
- But is this more ethical?

The debate:

- More honest or more deceptive?
- Pro: Respects privacy perception
- Con: Hides surveillance
- Which is worse?

Question: Is hiding your prediction capabilities ethical?

Lesson 1: Prediction Accuracy $\neq$ Ethical Use

Technical achievement:

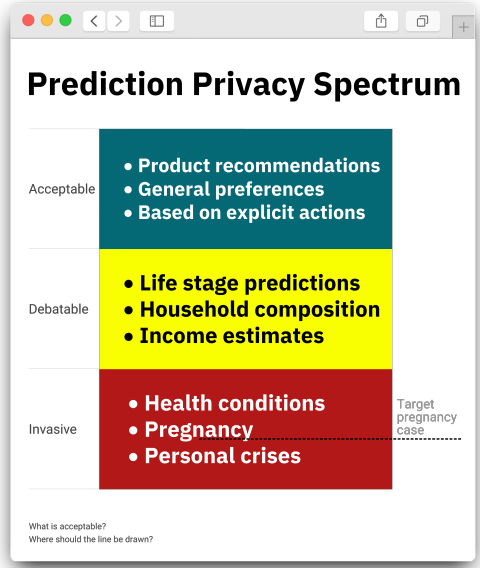
- Remarkable ML success
- High prediction accuracy
- Business value clear
- Sophisticated analytics
- Measurable ROI

But ethical questions:

- Just because we CAN predict...
- Should we?
- Who consents?
- What are boundaries?

Privacy Spectrum: Where to draw the line

- ✅ **Acceptable**
- ❓ **Debatable**
- ❌ **Invasive**



Lesson 2: Training Data Quality Matters

Why Target's approach worked:

- Baby registry = gold standard
  - Known pregnancy status
  - Known due dates
  - High-quality labels
- Thousands of examples
- Can look backward
  - See what they bought before registry
- Pattern detection possible

Potential problems:

- Selection bias
  - Only women who register
  - May miss patterns
- False positives
  - Gift buyers
  - Helping friends
- False negatives
  - Different shopping patterns
  - Online vs in-store

Lesson 3: Transparency vs Effectiveness (The tension)

Full camouflage:

- ✅ Maintains effectiveness
- ✅ No backlash
- ✅ Business value preserved
- ❌ Deceptive
- ❌ Hides surveillance
- ❌ Erodes autonomy

Example:

- Mix baby items with random products
- Pretend it's general catalog
- Hide the prediction

Question: Is there a middle ground? Should companies be required to disclose?

Would Target's approach be legal today?

GDPR requirements:

- **Lawful basis:**
  - Legitimate interest?
  - Or explicit consent needed?
- **Transparency:**
  - Right to know what data used
  - Right to know how processed
- **Purpose limitation:**
  - Can't use purchase data for unrelated purposes
- **Data minimization:**
  - Collect only what's needed

Ethics Discussion: Drawing the Line

1. ⚖️ Should there be a legal distinction between predicting:
   - Shopping preferences ("likes blue shirts") vs
   - Personal life circumstances ("is pregnant")?

   If yes, where exactly is the line?

2. ❓ Target's camouflage strategy:
   - More ethical (respects privacy perception)?
   - Less ethical (hides surveillance)?
   - Or just better PR?

Ethics Discussion: Drawing the Line

3. Who owns insights derived from your data?
   - You own transaction history
   - Does Target own the pregnancy prediction?

4. What would meaningful consent look like?
   - "We may infer pregnancy from purchases"?
   - Too specific to be practical?
   - Too vague to be meaningful?

Ethics Discussion: Consent & Control

5. $ Business vs privacy trade-off:
   - Target gains $600M/year
   - Customers lose privacy
   - Fair exchange?
   - Should customers be compensated?

6. Should regulations require:
   - Disclosure of predictive models?
   - Opt-out mechanisms?
   - Accuracy audits?
   - All of the above?

Modern Implications: This isn't just Target

Similar predictions today:

- Facebook: Relationship status
- Google: Income level
- Amazon: Financial stress
- Apple: Health conditions
- Insurance: Driving risk
- Universities: Dropout risk

The pattern:

- Behavioral data reveals everything
- Predictions getting more accurate
- Boundaries still unclear

Modern Implications

Emerging concerns:

- Health condition predictions
  - Insurance discrimination
- $ Financial distress
  - Predatory lending
- Academic performance
  - Admissions bias
- Relationship stability
  - Employment discrimination

Question: As predictions get better, do we need stronger protections?

Key Takeaways

1. **Transaction data is personal data:** Every purchase reveals life circumstances
2. **Patterns beat individual data points:** Combinations are powerfully predictive
3. **$ Business value is real:** $600M annually from one predictive model
4. **Accuracy $\neq$ Ethics:** Just because we can predict doesn't mean we should
5. **"Creep factor" matters:** Even accurate predictions can backfire
6. **Regulations catching up:** GDPR/CCPA now restrict these practices
7. **Lines are blurry:** Where's the boundary between personalization and invasion?

References

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